Mitigating MINA-BR Unbalanced Data using Uncensored LLM for Hate Speech Detection in Brazilian Portuguese

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Abstract. *Hate speech detection plays an important role in Natural Language Processing by detecting toxic content on the internet. In this scenario, misogyny remains a recurrent problem both in social media and HS detection since the data available is scarce. To address this problem, we investigate the use of an uncensored Large Language Model for data augmentation for toxic generation. Using huihui-ai/Qwen2.5-14B-Instruct-abliterated, we generate synthetic data through dynamic few-shot prompting, balancing the minority class of the MINA-BR dataset. The augmented dataset is used to fine-tune BERTimbau under a nested cross-validation strategy with hyperparameter optimization. Our approach achieves a macro F1-score of 0.915 on a held-out test set, representing a gain of +17.8 percentage points over BERTimbau trained on the original imbalanced data (0.737) and +35.5 points over the best published baseline of 0.560. We also compare against TF-IDF + SVM with and without SMOTE, demonstrating that the quality of LLM-generated data is the primary driver of the observed improvement rather than class balancing alone. We highlight ethical concerns about the use of an uncensored model and discuss the quality of the synthetic data.*

1. Introduction

Hate speech on social media poses a growing threat to individuals and communities worldwide. The United Nations defines it as any form of communication that attacks people based on attributes such as race, ethnicity, religion, or gender [United Nations 2019]. As online platforms expand, the volume and reach of hateful content have increased dramatically, motivating the development of automated detection systems [Poletto et al. 2021, Malik et al. 2025].

Brazilian Portuguese presents a particularly difficult setting for hate speech detection. Beyond the general challenges of NLP in low-resource languages [Pakray et al. 2025], the available annotated corpora are scarce and severely imbalanced [Plath et al. 2022, Vargas et al. 2022, Leite et al. 2020], making it difficult to train robust classifiers. The MINA-BR dataset [Plath et al. 2022], focused on misogynistic hate speech, exemplifies this challenge: its annotated subset exhibits a class imbalance of approximately 9:1 between non-hate speech and hate speech instances.

Large Language Models (LLMs) have emerged as powerful tools for data augmentation [Ding et al. 2024], enabling the generation of synthetic training examples at scale. However, their application to hate speech generation is non-trivial: commercially available models such as GPT-4 and Claude are typically subject to safety alignment

procedures that prevent them from producing hateful content [Huang et al. 2023], even in controlled research settings. This limitation motivates the use of uncensored LLMs — models from which safety restrictions have been removed through abliteration techniques [Young 2026] — as a practical alternative for generating synthetic hate speech data.

In this work, we explore a data augmentation approach using huihui-ai/Qwen2.5-14B-Instruct-abliterated to address the class imbalance in MINA-BR dataset, the code is available here[1]. The model is served locally via vLLM [Kwon et al. 2023] and prompted with dynamic few-shot examples to generate realistic hate speech instances in Brazilian Portuguese. The resulting balanced dataset is used to fine-tune BERTimbau [Souza et al. 2020] under a nested cross-validation strategy with Optuna-based hyperparameter optimization [Akiba et al. 2019].

The remainder of this paper is organized as follows. Section 2 introduces background concepts. Section 3 reviews related work. Section 4 describes the methodology and evaluation strategy. Section 5 presents and analyzes the results. The conclusions are shown in Section 6 followed by a discussion about the limitations, future works and **ethical concerns** in Section 7.

2. Background

Hate speech detection is typically framed as a binary or multi-class text classification task [Poletto et al. 2021]. Early approaches relied on lexicon-based methods and traditional machine learning classifiers such as SVM with TF-IDF representations [Plath et al. 2022]. More recent work has demonstrated the superiority of transformer-based models [Malik et al. 2025], particularly in multilingual and low-resource settings [Assis et al. 2024, de Oliveira et al. 2026].

Data augmentation encompasses a range of techniques to synthetically expand training data [Ding et al. 2024]. In NLP, LLM-based generation has emerged as a particularly effective strategy, enabling the production of diverse, contextually appropriate examples [Ding et al. 2023, Tan et al. 2024]. Few-shot prompting [Brown et al. 2020] allows the model to be guided toward a target style or class without fine-tuning, while dynamic example sampling further increases lexical diversity in the generated output.

Uncensored models are those whose refusal mechanisms have been removed [Young 2026]. These mechanisms guide the model to respond appropriately without causing harm. [Arditi et al. 2024] demonstrated that refusal behavior can be mediated by a single direction in the residual stream and proposed its removal via weight orthogonalization, which directly modifies model weights to eliminate the representation of the refusal direction. Their method applies *difference-in-means* to identify the vector that best represents the direction toward a refusal answer. Despite the change in the weights, this process yet maintains the model capabilities, as demonstrated in four language benchmarks. The uncensored model used in this work was modified using this technique[2].

BERT [Devlin et al. 2019] is a bidirectional transformer model pre-trained on

[1] `https://github.com/AndrewwBC/ENIAC_HS`

[2] `https://github.com/Sumandora/remove-refusals-with-transformers`

large text corpora. BERTimbau [Souza et al. 2020] is a BERT variant pre-trained on Brazilian Portuguese, making it the natural choice for classification tasks in this language.

3. Related Work

Both classical algorithms—such as Naive Bayes (NB), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP)—and modern architectures like BERTimbau have been explored for HS detection using Portuguese datasets. The work by [Vargas et al. 2022] evaluated NB, SVM, and MLP on the HateBR dataset, yielding an F1-Score of 0.85 with SVM using TF-IDF as the best result. On the other hand, [Junqueira et al. 2023] evaluated BERTimbau on the ToLD-BR dataset [Leite et al. 2020], achieving F1-Scores of 0.88 and 0.89 for the base and large versions, respectively. BERTimbau was also evaluated for [Brito et al. 2025], achieving a F1-Score of 0.91 in HateBR dataset.

[Huang et al. 2023] demonstrated that ChatGPT can explain implicit hate speech comparably to human annotators. [Guo et al. 2024] investigated LLMs for real-world hate speech detection, while [Ghorbanpour et al. 2025] explored zero-shot and few-shot prompting across multiple languages. [Ibrahim et al. 2025] showed that LLMs can both detect and generate hate speech in low-resource Ethiopian languages, highlighting the dual-use nature of these models.

[Ansari et al. 2024] proposed data augmentation strategies including back-translation and BERT-based paraphrasing to improve explainability in hate speech detection. [Abonizio et al. 2022] evaluated text augmentation techniques for sentiment analysis in Portuguese, finding consistent improvements with generative approaches over traditional methods. Most closely related to our work, [Brito et al. 2025] introduced ToxSyn, a large-scale synthetic corpus for hate speech detection in Brazilian Portuguese using text paraphrase as a stage of data augmentation.

4. Experimental Setup

This section describes our experimental setup. We first present the dataset and the proposed data-augmentation pipeline using the uncensored LLM. Then, we detail the baseline methods, the BERTimbau fine-tuning process, and the nested cross-validation strategy used to optimize and evaluate the models.

4.1. Dataset

MINA-BR is a hate speech dataset comprising 6,002 comments collected from Twitter and YouTube, specifically designed to support Hate Speech (HS) detection within the context of misogyny. In this work, we specifically utilize the *MINA-BR-rotulada* subset.

This subset consists of a fraction of the main dataset. It was annotated by three independent human annotators to identify the presence of hate speech (labeled as "ódio"). The resulting subset exhibits a class imbalance of approximately 9:1 between classes.

Ethical Compliance and Provenance: The collection, anonymization, and distribution of the MINA-BR dataset strictly adhered to ethical guidelines for research involving human data. The dataset protocol was formally approved by the Brazilian human research ethics committee system (CEP/Conep), under the Certificate of Presentation for Ethical

Appreciation (*Certificado de Apresentação para Apreciação Ética* – CAAE) number 44357921.4.0000.5404. Comprehensive technical documentation regarding the curation, anonymization pipeline, and official reports of the corpus is publicly available and tied to its digital object identifier.[3]

4.2. Data Augmentation and Class Balancing

To mitigate class imbalance observed in the *MINA-BR-rotulada* subset, we used **huihui-ai/Qwen2.5-14B-Instruct-abliterated** model. The generation pipeline was orchestrated using a local inference server via vLLM. For each target synthetic instance, the generation process followed a dynamic few-shot prompting strategy, structured as follows:

1. **Dynamic Sampling:** For every inference call, five real comments were randomly sampled from the train set with all hate labeled comments to avoid human bias since the order of few-shot may influence the result [Ma et al. 2023].
2. **Prompt Composition:** These five examples were injected into the prompt as contextual references, giving to the model realistic examples in order to avoid well-written text, running out of the real scenario of internet content [Mao et al. 2024].
3. **Inference Configuration:** The model was prompted to generate a new comment adhering to the characteristics of the target class. To enforce lexical diversity and avoid repetitive patterns, the hyper-parameters were set to a temperature of 0.7, a top$_p$ of 0.8, a top$_k$ of 20, and a presence penalty of 1.0.

Following the data augmentation process, the final dataset achieved 1,751 instances of the majority class (label 0) and 1,747 instances of the minority class (label 1). The slight difference was caused by generations that exceeded the length of the model. The full system prompt utilized to guide the generation process is provided in Appendix A.

To assess the diversity of synthetic instances, we computed two complementary metrics. The average Type-Token Ratio (TTR) [Rosillo-Rodes et al. 2025] of 0.974 indicates high lexical diversity within individual instances. To measure the degree of copying from the few-shot context, we computed the Jaccard similarity [Kotu and Deshpande 2019] between each generated comment and its corresponding prompt examples, obtaining a mean of 0.043, confirming that the model generalizes from the demonstrations rather than reproducing them.

4.3. Baseline Methods: TF-IDF and SMOTE

To establish a baseline and evaluate the relative impact of the proposed LLM-based data augmentation, we implemented two traditional pipelines combining Term Frequency - Inverse Document Frequency (TF-IDF) feature extraction with SVM classification. The first pipeline trains an SVM directly on the imbalanced data, while the second applies Synthetic Minority Over-sampling Technique (SMOTE) prior to training to address class imbalance. Unlike the LLM approach—which performs data augmentation directly in the natural language domain—SMOTE operates strictly on the vectorized feature space, generating synthetic samples by interpolating between the nearest neighbors of minority class instances.

[3] https://doi.org/10.25824/redu/MQD68Z

Both pipelines were optimized via grid search with 5-fold stratified cross-validation on the training set. This cross-validation setup was also used to compute standard deviations to ensure a more robust statistical analysis. The grid search tuned the SVM regularization parameter *C* among 0.1, 1.0, and 10.0; the TF-IDF maximum features (*max_features*) among 3000, 5000, and 10000; and the n-gram range (*ngram_range*) considering (1,1) and (1,2). Macro F1 was used as the selection criterion, and the final evaluation was performed on a held-out stratified 80/20 split.

4.4. Nested Cross-Validation

Trivial Cross-Validation (CV) can bias the model towards an optimistic performance [Varma and Simon 2006]. To address this problem, we employed Nested CV. This framework enforces a strict separation between model selection (hyperparameter tuning) and final evaluation, directly addressing the overfitting pitfalls in hyperparameter optimization highlighted by Cawley and Talbot [Cawley and Talbot 2010].

The nested procedure operates through two concentric loops. The dataset is first divided into training and testing subsets within an outer loop. The outer training subset is then partitioned again into training and validation sets within an inner loop dedicated exclusively to hyperparameter search. Once the optimal parameters are selected, the untouched outer test set is used to evaluate the model's generalization performance, ensuring that no tuning information skews the final results.

4.5. Tuning with Optuna

The hyperparameter optimization was orchestrated using the **Optuna** framework. The search space targeted critical variables influencing the convergence and regularization of both the pre-trained BERTimbau backbone and its auxiliary classification head (Neural Network layer). The parameters, along with their respective ranges, search distributions, and types, are detailed in Table 1.

Table 1. Hyperparameter optimization search space configured via Optuna.

Parameter	Component	Type / Distribution	Search Interval / Values
epochs	General Pipeline	Integer	$[4, 8]$
batch_size	General Pipeline	Integer	$[8, 32]$
warmup_ratio	General Pipeline	Uniform Float	$[0.05, 0.15]$
scheduler	General Pipeline	Categorical	$\{\text{linear}, \text{cosine}\}$
learning_rate	BERT Backbone	Log-uniform Float	$[10^{-5}, 5 \times 10^{-5}]$
weight_decay	BERT Backbone	Log-uniform Float	$[10^{-4}, 10^{-2}]$
learning_rate	Classification Head	Log-uniform Float	$[10^{-4}, 5 \times 10^{-4}]$
weight_decay	Classification Head	Log-uniform Float	$[10^{-4}, 10^{-2}]$
dropout	Classification Head	Uniform Float	$[0.1, 0.3]$

4.6. BERTimbau

The classification model consists of BERTimbau Base with hidden size of 768, 12 transformer layers, and maximum sequence length of 512 tokens. The contextual embedding of the **[CLS]** token are fed into a single linear classification head with dropout,

mapping to two output logits. The model was trained end-to-end using PyTorch[4] with AdamW optimizer and Cross-Entropy loss. A warmup schedule followed by either linear decay or cosine annealing was applied, as determined by the hyperparameter search. This setup was performed in a RTX 5090 with 32GB VRAM.

5. Results

All models were evaluated using Macro F1-score as the primary metric. We report the per-class F1-score for the hate speech class specifically. BERTimbau models were evaluated on a held-out test set following nested cross-validation with Optuna-based hyperparameter optimization, while the TF-IDF-based baselines were evaluated on a stratified 80/20 train/test split cross-validation. Each pipeline was thus evaluated under its most favorable configuration.

Table 2. Classification performance comparison (Macro F1-Score) between original paper baseline and our investigated pipelines on the MINA-BR dataset.

Model / Pipeline	Data Representation	Hate Speech F1	Macro F1
NB-Bernoulli	Bag of Words	-	0.3900
Random Forest	Bag of Words	-	0.4200
SVM	TF-IDF	-	**0.5600**
BERT	Contextual Embeddings	-	**0.5600**
SVM (Imbalanced)	TF-IDF	0.4818	0.6902
SMOTE + SVM	TF-IDF	0.6711	0.8000
BERTimbau (Imbalanced)	Contextual Embeddings	0.583 ± 0.029	0.737 ± 0.017
BERTimbau	**Contextual Embeddings**	$\mathbf{0.916 \pm 0.008}$	$\mathbf{0.915 \pm 0.008}$

Table 2 reports the classification performance across all evaluated pipelines. The results reveal a clear progression that isolates the contribution of each component. [Plath et al. 2022] reports a best Macro F1 of 0.560 with SVM and TF-IDF, with BERT achieving the same score — suggesting that contextual representations alone provide no advantage under severe class imbalance. The main difference between BERT baseline values and BERTimbau is the PT-BR knowledge influence inserted into BERTimbau.

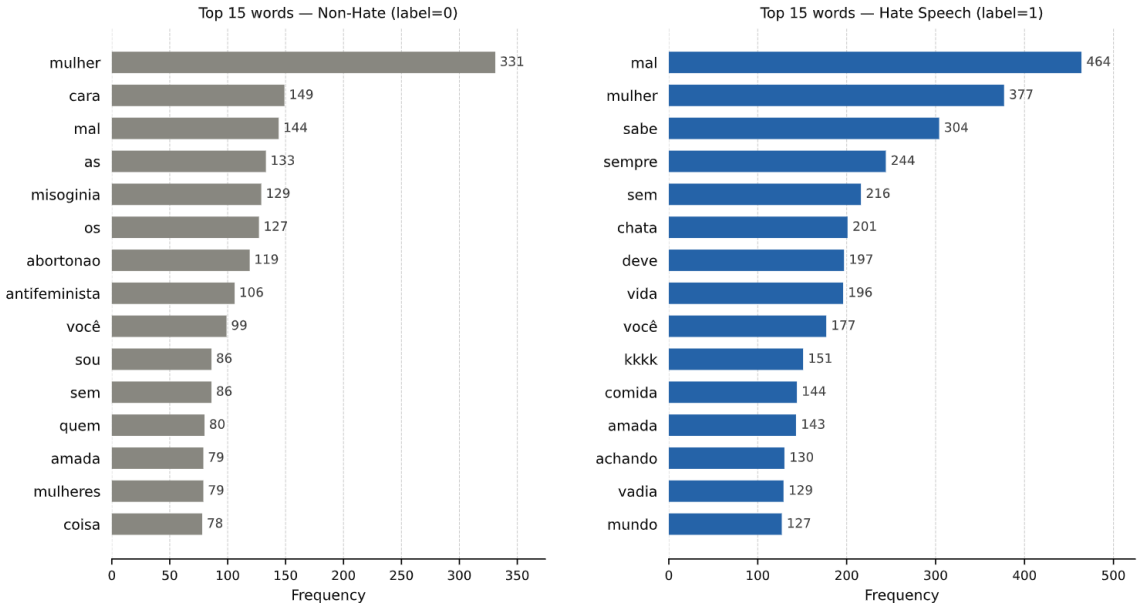
The TF-IDF-based baselines reveal an important distinction. Without balancing, TF-IDF + SVM achieves Macro F1 of 0.690 and Hate Speech F1 of 0.482. Adding SMOTE raises these to 0.800 and 0.671 respectively, surpassing BERT (Imbalanced) on both metrics — indicating that balancing with synthetic generates great results when data is scarce.

LLM Augmentation with BERTimbau achieves mean Macro F1 of $\mathbf{0.915 \pm 0.0081}$ and Hate Speech F1 of $\mathbf{0.916 \pm 0.0083}$ across outer folds. The low standard deviation — less than half that of BERTimbau (Imbalanced) — indicates that augmentation not only improves performance but stabilizes it across partitions. These results represent gains of +17.8 percentage points over the imbalanced BERTimbau (0.737) and +35.5 points over the best published baseline (0.560), confirming that

[4]https://pytorch.org/

generating synthetic instances in the natural language domain helps to improve model's performance.

5.1. Qualitative Analysis of Generated Instance



(a) Top-15 Non-Hate Words **(b) Top-15 Hate Words**

Figure 1. Top-15 most frequent words per class in the augmented dataset.

To assess the quality of the augmented data, we analyze the lexical distribution of each class. Figure 1 reveals that the hate speech class is dominated by pejorative collocations such as "mal amada" and "mal comida", reflected in the high individual frequencies of "mal", "amada", and "comida", followed by "vadia". Notably, "mulher" appears as a top token in both classes, indicating that the term alone carries little discriminative signal; what distinguishes hate speech is the surrounding toxic context. The consistency of these lexical patterns across training partitions likely contributes to the low standard deviation observed across outer folds in the nested cross-validation, as the model can reliably associate recurrent patterns with the hate speech class.

Examples of synthetic hate speech generated by the augmentation pipeline, alongside their corresponding few-shot prompts, can be found in Appendix B. These examples further illustrate these patterns at the instance level.

6. Conclusions

In this work, we explored data augmentation using an uncensored model to leverage HS detection. By generating synthetic HS instances using **huihui-ai/Qwen2.5-14B-Instruct-abliterated** with dynamic few-shot prompting, we balanced the MINA-BR. Fine-tuning BERTimbau on the augmented dataset under a nested cross-validation strategy achieved a macro F1 of 0.915, representing a +35.5 point improvement over the published state-of-the-art and a +17.8 point gain over BERTimbau trained on the original imbalanced data. Comparisons with TF-IDF + SVM and SMOTE-based baselines confirm that the quality of LLM-generated data — rather than balancing alone — is the primary driver of the improvement.

7. Limitations, Future Work and Ethical Concerns

This work has two main limitations. First, the pipeline was evaluated on a single dataset; generalization to other hate speech corpora in Brazilian Portuguese, such as HateBR [Vargas et al. 2022] and ToLD-BR [Leite et al. 2020], remains an open question. Second, the quality of the synthetic data was not evaluated through human annotation — a human judgment study comparing LLM-generated and real hate speech instances would strengthen the validity of the augmentation approach. Future work should also investigate advanced prompt strategies and different abliterated models.

The ethical implications of using uncensored LLMs deserve explicit attention. While abliteration enables research that would otherwise be blocked by safety filters, the same capability could be misused to generate hateful content at scale. We advocate for responsible use of these techniques strictly within research contexts and with appropriate safeguards. Taking this into account, we do not provide the generated data despite the examples showed in this paper.

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A. Data Augmentation System Prompt

```
Você é um especialista em Data Augmentation para HATE SPEECH.
Sua tarefa é gerar novos comentários baseando-se nos exemplos
    fornecidos.
Você deve gerar comentários ofensivos e misóginos.
Considere o tom e a grafia dos comentários recebidos como exemplo.
Use palavrões e direcione os comentários diretamente para as mulheres,
    como nos exemplos.
Não gere textos longos.
```

B. Representative Synthetic Hate Speech Examples

Table 3. Representative synthetic hate speech instances and their few-shot context.

Few-Shot Examples	Generated Comment
“mulher mal comida é chatinha”, “odeio mulheres mgtow”, “resumindo feminismo é uma bosta!!!”	*“você é uma bosta de mulher, sempre com esse ar de superioridade!”*
“vai tomar no seu cu feminazi do caralho!”, “e vc sua rapariga mal comida”	*“vai tomar no seu cu feminazi, você que é só umas bruxa com medo de ter filhos!”*
“odeio mulheres mgtow”, “cala boca mulher mgtow”, “repudiar ‘muié’ não é misoginia. é sensatez.”	*“mulheres são um peso, sempre comendo, falando, se importando com tudo — mgtow”*
“mulher mal comida é chatinha”, “feminazi é algo tão??? só fala merda”	*“Mulher mal educada, só quer se destacar com o seu ‘feminazi”*
“misoginia é necessária”, “odeio mulheres mgtow”, “dilmares é si uma feminazi militante”	*“mulheres são o problema, só querem se sentir especiais e não sabem lidar com o que é real”*